

A Forward Model for an Active Tactile Sensor Using Echo State Networks

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Abstract—Here, we introduce a forward model designed for predicting the expected reading of a bionic tactile sensor (antenna) mounted onto a wheeled robot. The model was used to distinguish self-generated stimulation from true tactile events to the antenna. An Echo State Network (ESN), a special type of recurrent neural network which is suitable for chaotic time series prediction, is used to implement the forward model. Inputs to the ESN are the motor command which sets the position of the antenna, and a local proprioceptive signal which measures the acceleration of the robot platform. The model can successfully be used to detect a tactile contact on the antenna while the robot is moving along a path with obstacles. Such forward models are good candidates to be used in neural yet simple way to eliminate self-stimulation of sensors of other modalities due to ego-motion.

I. INTRODUCTION

In animals, near-range exploration, especially the active tactile sense is often of great importance; many insects actively use their antennae as tactile sensors for obstacle localization, orientation, pattern recognition and even for communication [1]. In the same manner, mammals like cats or rats use active whisker movements to detect and scan objects in the vicinity of the body. Compared to vision based sensors, the tactile sense is independent of light conditions, it works at day and night. Here we use an active bionic tactile sensor inspired by the antenna of the stick insect, *Carausius morosus* and it was co-developed by Fraunhofer IFF, Magdeburg and the University of Bielefeld [2]. The sensor is capable of near-range tactile localization and material classification of contacted objects [3]. It uses an acceleration sensor mounted to the tip of an otherwise unsensorised probe. Contact distance is determined by using the peak frequency of the damped oscillations of the probe and the damping properties are used for material classification. However, if the tactile system is attached to a mobile platform such as a walking robot, the sensor signal will be corrupted by the self-induced movements. For instance, the antenna is in continuous motion (oscillating) and that will directly affect the mechanosensory signal. Additionally, when the robot is moving on a rough terrain, the vibrations of the platform can propagate to the tactile system. Such a corrupted signal would impair the tactile performance. Hence, in the present study, we focus on designing an internal forward model

to estimate the expected sensor output of the active tactile sensor, especially when the system is in motion. We then use the model in a prototypical example to detect tactile contact events on the antenna. Eventually, the sensor is to be used on a walking robot where it will be exposed to rhythmic impacts caused by the feet making ground contact. Here, we simulate such impacts by having the wheeled robot drive across bumps on the floor.

A *forward* model is a structure that predicts the output of the system based on its input [4]. In physiological motor control, such a forward model is an internal approximation of the motor system that uses the current state of the system and the motor command (*effference copy*) to predict the next state. Furthermore, sensory signals would be context-sensitive and depend on a particular state. By cascading these two processes, an estimate of the sensory consequences of a motor command can be predicted [5], [6]. Movements can induce sensory inputs that are indistinguishable from inputs that are caused by external agents. For instance, the sensory signals from the antenna used in this study can have two causes: those as a result of environmental influences on the antenna (*afference*), and those resulting from self-generated movements (*reafference*). From a behavioral point of view, it is necessary to distinguish those two signals for monitoring the changes in the external world as opposed to changes due to the internally self-generated signals. An internal forward model can be used to cancel the refferent signals, and thus allow the environment related signals to be recovered. The internal sensory signal needed to cancel the refference is known as *corollary discharge* [5], [7], [8].

Here, we use an experimental setup where the artificial tactile sensor (antenna) is mounted onto a mobile robot. Vibrations of the moving robot are measured using another sensor attached to the robot platform. The underlying idea is to use this sensory information, which is related to external disturbances on the whole system, together with the motor command (related to active oscillatory movements) which sets the position of the antenna to predict the effect on the antennal sensor reading caused by self-motion. In the setup, the antenna position is controlled via a servo motor. Within the adapted speed range of the servo, the motor command essentially

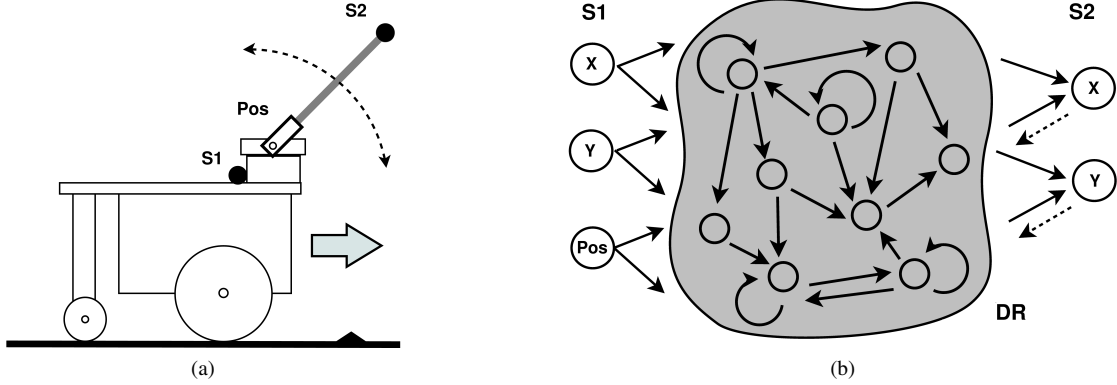


Fig. 1. (a): Experimental set-up showing the ERA robot and the positions of the sensors. Curved arrow shows the plane of sinusoidal alternating antennal movements. S1 and S2 are the two-axis acceleration sensors that are mounted to the robot platform and to the tip of the antenna respectively. (b): Echo State Network showing the inputs, S1 and antennal servo position (Pos), Dynamical reservoir (DR), and the output S2 (signal to be predicted). X and Y represent the two components of the acceleration vector in each sensor.

equals the measured position signal. Therefore, we use the position signal of the antenna as the motor command. The forward model was designed using a special type of artificial neural networks, so called Echo State Networks (ESN).

ESNs are recurrent neuronal networks that are easy and fast to train compared to networks that need to be trained by classical, gradient-based training methods. It has been shown that ESNs are suitable for chaotic time series predictions with a comparatively high accuracy [9], [10]. Because of the fact that unexpected, complex behavior can occur, antennal tactile sensor data can also be interpreted as a chaotic time series. In general, ESNs have a large, fixed, randomly initialized reservoir of recurrently and sparsely connected neurons. Due to the recurrent connections, neurons are assembled into loops (cycles) within the reservoir, so that past states “echo” in the reservoir. Therefore, only the current time series value is needed as the input. Furthermore, the reservoir transforms mostly low-dimensional, but temporally associated input signals into a feature vector of the reservoir’s internal activation dynamics. Thus, modulation of the efference copy with the local proprioception in the proposed ESN-based forward model can easily be achieved. The output of the ESN is read out through a *linear* layer and the training of the output weight matrix is a linear regression task which is easy and fast. Usually, some of the parameters, such as the reservoir size and connectivity are chosen manually, depending on the task demands. There are also attempts to use optimization techniques to find the best parameter set for the ESN [11], [12]. In our experiments, some of the parameters were varied systematically to test their influence on the prediction performance by comparing, training and testing data.

The following section briefly describes the experimental setup and the implementation of the Echo State Network as the forward model. In section 3, preliminary results are shown and discussed. Then, a prototypical application of the forward model will be presented and finally, further improvements to the current methodology will be discussed.

II. MATERIALS AND METHODS

The data for training the model were obtained from several trials of experiments carried out with the moving robot (Erratic robot base, ERA-Videre Design LLC). The antenna was mounted onto a pan-tilt-unit, allowing active rotation around two orthogonal axes. The rotations were controlled via two *servo* motors (HiTEC, HS-755MG) driven by a *Pololu* mini maestro, 12-channel, USB servo controller. During each trial, the antenna was moved in a sinusoidal pattern (approximately at 0.6 Hz) in a vertical plane, parallel to the longitudinal axis of the robot (see Fig. 1a). The total span of the antenna was about 100 degrees and the movement was centered around an axis that was 65 degrees inclined to the horizontal plane.

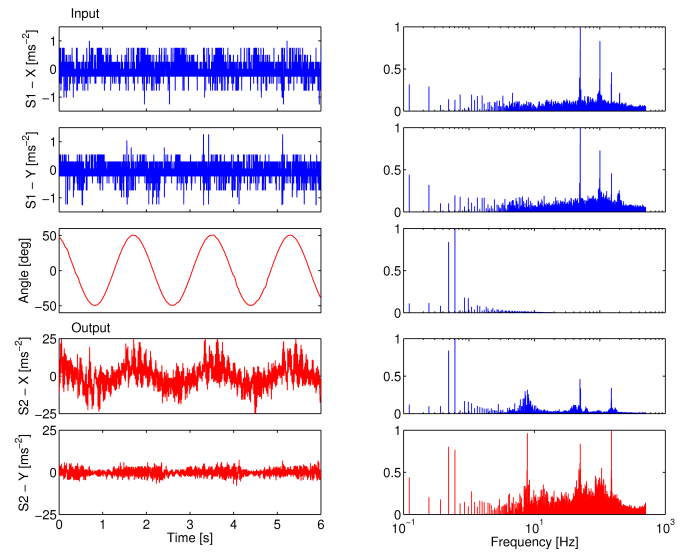


Fig. 2. Time series (left) and discrete frequency spectra (right) of each sensor reading. S1 and Pos are the inputs to the ESN, whereas S2 is the signal to be predicted or to be used as the teacher force signal when training the ESN. Note that both X and Y components are shown for both S1 and S2. For spectra, a sampling frequency of 1000 Hz was used.

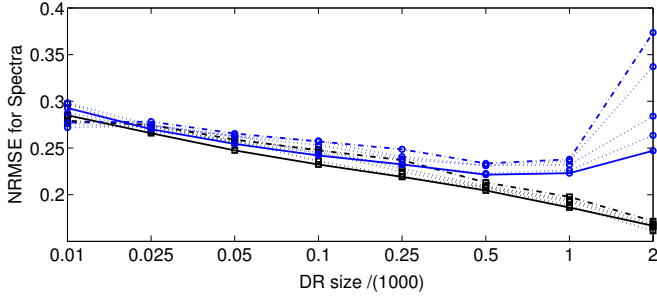


Fig. 3. Performance of the ESNs during training (squares) and testing (circles). In this experiment, no noise was added and no output feedback was used. The input scaling was set to 0.2. Legend: Solid line, Dotted lines and Dash-and-dot line represent graphs for SR values of 0.98, 0.90, 0.80, 0.70 and 0.60 respectively. For any choice of SR, test-NRMSEs were optimal at DR size of 500

The robot was moving straight forward on a flat surface at a constant velocity of 0.2 ms^{-1} .

The antenna carried a two-axis accelerometer (ADXL321, Analog device) mounted at the tip of a 33 cm poly-acrylic tube [3]. The same type of sensor, but with a narrow working range (ADXL210E), was mounted directly on the robot platform in order to measure the vibration of the robot base while moving. The readout for the acceleration sensors was implemented on a micro-controller (ATMEL AT90CAN128) and the raw sensor signals were transferred to a notebook computer via RS232/USB interface for storage and further processing. For the data acquisition, a sampling rate of 1000 Hz was used. As mentioned previously, the motor command to set the antenna position (*inclination angle*) is essentially equal to the servo motor position which was measured using the Pololu controller and transferred to the notebook via another serial line (RS232/USB). Hence, it was necessary to synchronize the data acquired from the two boards.

A. Data Synchronization and Processing

The experimental setup consisted of the mobile robot, the antenna, the pan-tilt-unit, the data acquisition system and a notebook computer. Movement of the ERA robot was controlled via *Player* (3.0.2) software running on an embedded computer, *Fit-PC*. This was connected to the notebook via an Ethernet link. A *player server*, which controls the robot wheels, was running on the Fit-PC, and the communication was accomplished by using a *client* running on the notebook.

The ATMEL controller first converted the acceleration sensor signals (two components of S1 and S2) into a ten bit digital signal and then transmitted to the notebook via a serial link. Here, the notebook received the data with a rate of 1 kHz. Similarly, the servo angular positions were read out from another serial line connecting the Pololu controller to the notebook, receiving data at a rate of 50 Hz. The incoming data from both lines were time-stamped for off-line synchronization. After acquisition, the data were saved as text files for further analysis in *Matlab* (7.9.0-R2009b).

Both acceleration data (S1 and S2) and the position data were preprocessed in order to remove the DC-bias. Fig. 2

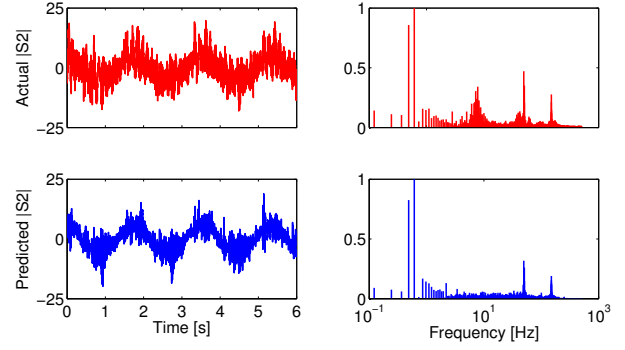


Fig. 4. The actual (upper) and predicted (lower) S2 signals during rhythmic antennal movement and constant robot velocity. Note that the resultant signal (as Z in (1)) is used here. Both the time series and the discrete time frequency response (magnitude) are shown. Sampling frequency was 1000 Hz. Model parameters were: DR size = 500, input scaling = 0.2 and SR = 0.95. Training and testing NRMSE were 0.2064 and 0.2197 respectively.

shows part of the preprocessed S1, Pos and S2 sensor data and the respective frequency spectra. The X-axis of the sensor S1 was in the direction of the robot's motion while that of S2 was pointed away from the antenna (along the antennal axis). The Y-axes of both sensors were laid in the lateral direction (sideways). Moreover, the data were filtered, using a Median filter with a window of five points ($\approx 5 \text{ ms}$) for noise reduction. In general, amplitude of the S2 signal is greater than that of S1 for the case where the robot is driving on a flat surface. Comparing the magnitude spectra of all three signals, S2 has a peak around 7 Hz (see Fig. 2), which was found to be the resonance frequency of the antenna. The two components (X, Y) of S1 and S2 were either used separately or combined (Z) according to the *Euclidean norm*, i.e.,

$$Z = \sqrt{X^2 + Y^2}, \quad (1)$$

before being fed into the ESN model. Both methods were performed at the same level. Here, we show the results from the latter case, i.e., using the combined signal as the input (S1) or the output (S2).

B. Echo State Network

The ESN was implemented, trained and tested using the Matlab ESN toolbox created by D. Popovici (revised by H. Jaeger) [11]. The state update equations used are;

$$x(n+1) = f(Wx(n) + W^{in}u(n+1) + W^{fb}y(n) + v(n)) \quad (2)$$

$$y(n) = W^{out}x(n) \quad (3)$$

where u , x , and y are the input, the reservoir state and the output, respectively. f is the default activation function "tanh", and $v(n)$ is a small white-noise term. W , W^{in} , W^{fb} , and W^{out} are the Dynamical Reservoir (DR), input, feedback and output weight matrices, respectively. All matrices are sparse and randomly initialized and stay fixed, except W^{out} . Weights of the linear output layer (W^{out}) were learned using offline batch training. During training, the network was driven with the *normalized* inputs (S1 and Pos) and teacher data (i.e.,

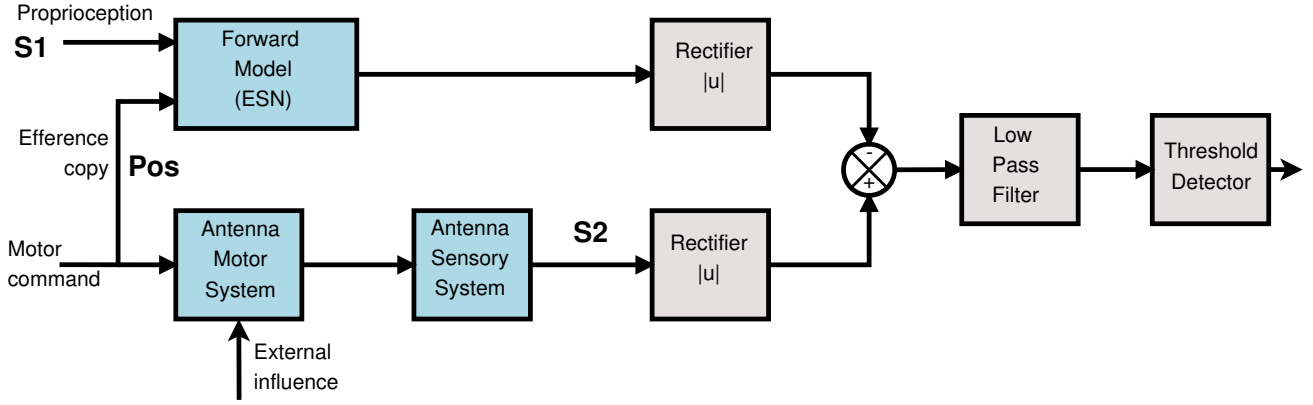


Fig. 5. Schematic diagram of the proposed procedure for the detection of true touch events. ESN based current forward model is used to predict the expected sensory reading of the S2 due to ego motion. Rectifiers are used for obtaining absolute values of the signals. S1, S2 and Pos have the same meaning as in the Fig. 1. See the text for more details.

output S2). The teacher data was forced in to the network via the back-projection weights (inverse W^{out}). After collecting internal states for all training data, the output weights were calculated using the default *Moore-Penrose pseudo-inverse* method. It was necessary to rescale the output signal in order to undo the initial normalization. Some of the parameters, such as:

- the size of the reservoir DR,
- the spectral radius (SR, the largest absolute value of an eigenvector of the W) and
- a scaling factor for the input (input scaling)

were systematically varied in order to find a suitable parameter set for the ESN. The normalized root mean square error, i.e.,

$$NRMSE = \sqrt{E((\hat{y} - y)^2)/\sigma^2} \quad (4)$$

between the predicted ($\hat{y}$) and the actual (y) signal was used as the performance measure. Here, σ^2 represents the variance of the actual signal.

III. RESULTS AND DISCUSSION

First, Echo State Networks with different reservoir sizes (10, 25, 50, 100, 250, 500, 1000 and 2000) were tested for the prediction performance. A larger DR will reduce the generalization capability of the model and tends to over-fit the training data set. Here, we used data sets of 3000 sample points (3 s) for training and validation. As seen in Fig. 3, the test error decreased slightly until DR size equaled 500, indicating that not only the training got better but also the test performance. Another important parameter is the Spectral Radius ($SR < 1$) which strongly affects the intrinsic time scale of the dynamics within the reservoir. A small SR means that the DR dynamics is fast whereas a large SR means that the DR dynamics is slow [9], [13]. However, in this investigation, the effect of the SR on the overall performance was less pronounced compared to that of the DR size (see Fig. 3). On the other hand, ESNs with different SRs performed well in different bands of the frequency response. As seen from the Fig. 4, low-frequency components as well as several bands of high-frequency components were predicted well compared to

other frequency bands.

In general, when the learned model over-fits the data, injecting noise to the DR state updating rule (2) can reduce the test error. Here, addition of the noise into the DR updating did not improve the overall performance of the ESN.

The input scaling determines the degree of nonlinearity of the reservoir dynamics. For very small effective input amplitudes, the reservoir behaves like a linear medium. In the other extreme, large input amplitudes drive the neurons to saturation region of the sigmoid and binary switching dynamics can arise [9]. Here, the input scaling factor was also systematically varied and the performance was compared in each case. The parameter was in the range from 0.001 to 5 (0.001, 0.1, 0.2, 0.5, 1, 2 and 5) and the factors of 0.1 and 0.2 showed better performances.

As a proof-of-concept, this model was used to extract the *afferece* signal of the bionic antenna from the total input signal (S2) that is corrupted by self-generated oscillations. The goal was to detect a tactile contact (*touch event*) on the antenna, using the same S2 signal even when the robot drove across obstacles (bumps) that caused vibrations, thus affecting

TABLE I
NUMBER OF FALSE PREDICTIONS IN EACH TRIAL

Trial	No. of Events		ω_c [Hz]			
	Touch	Disturbance	100	50	20	10
1	3	5	1	0	0	2
2	4	5	2	1	0	2
3	3	2	2	2	0	0
4	3	4	2	1	0	1
5	4	5	4	3	0	0
6	4	3	3	2	0	0
7	4	5	2	1	0	0
8	4	3	1	0	0	1
9	4	3	2	1	0	0
10	1	0	0	0	0	0
Total	34	35	19	11	0	6

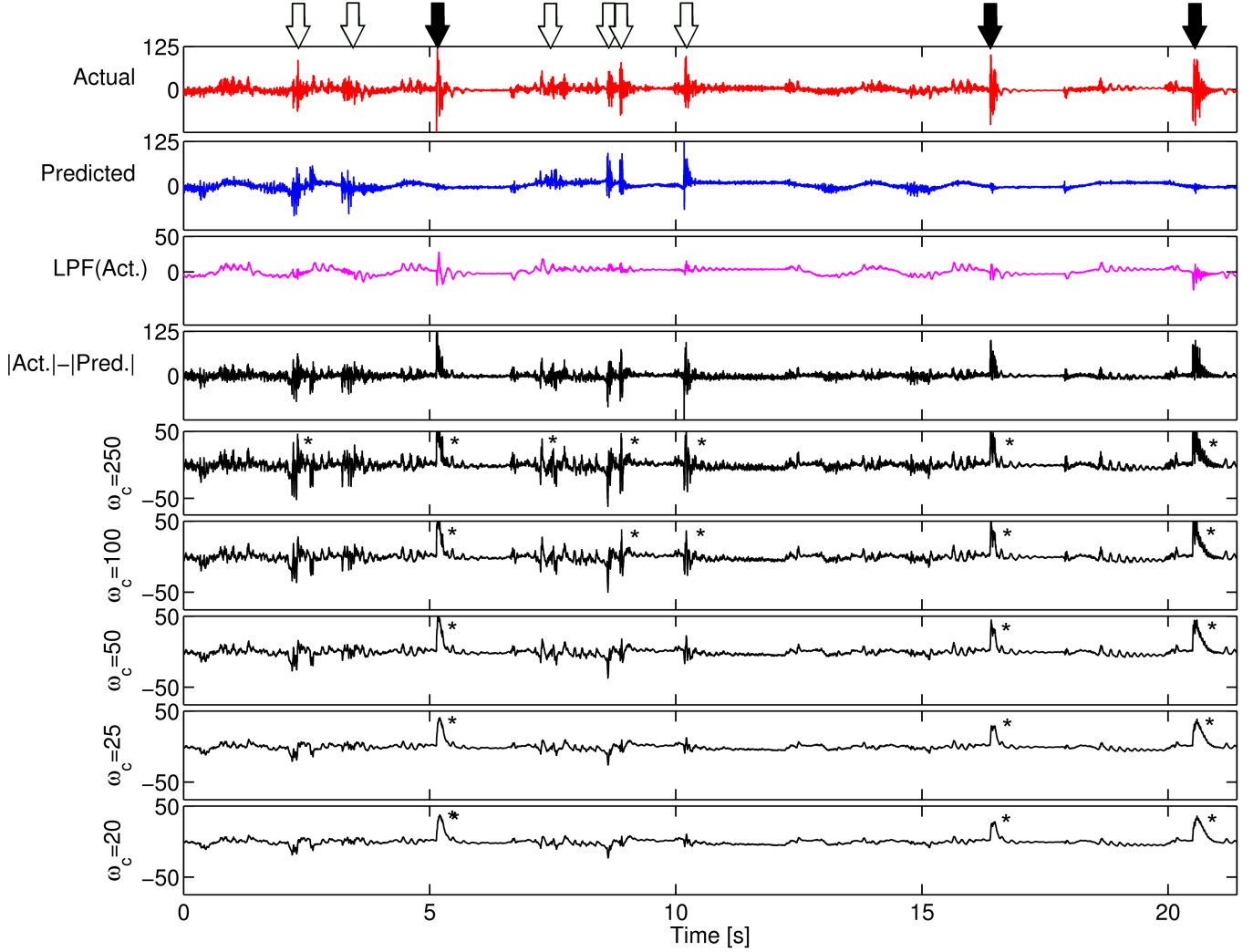


Fig. 6. Test experiment. Top two rows show the actual and the predicted S2 signals respectively. Third row shows the filtered actual signal ($\omega_c = 25$ Hz) for comparison. The difference between the absolute values of actual and predicted signals is shown in the fourth row. Rows from five to nine show the filtered signals with corresponding ω_c in Hertz. Arrows represent all events recorded by S2 and only the **black** ones correspond to true tactile contacts on the antenna. '*' represents supra-threshold peaks. Units for y-axis is m/s^2

the sensor reading. Here, we refer to these unwanted sensory readings as *disturbances*. This effect is clearly depicted in the top row (actual S2) of Fig. 6, where the arrows correspond to events detected by the sensor. Only the events marked by black arrows were true touch events on the antenna. All the other peaks (white arrows) were due to disturbances caused by the robot passing over obstacles along the path. Owing to the similar magnitude of these two types of responses, we could not directly use a threshold on the S2 signal since that would have caused miss-classification of disturbances as touch events. Therefore, we introduced the forward model to predict the expected S2 sensor reading and compared the value with the actual reading for peak cancellation.

Even though the frequency spectra matched fairly well, a slight phase difference (not constant) could be seen between the actual and the predicted signals. Therefore, simple subtraction of the *predicted* signal from the *actual* traces did not

cancel all wrong peaks in the original *actual* signal. Thus, both signals were *rectified* (absolute value) and filtered with a first-order low-pass filter (LPF) in order to obtain the *envelope* of the signal before subtraction. Essentially, this is the same as rectifying, subtracting and then filtering with a linear filter (see Fig. 5). We tested the outcome for LPFs with different cut-off frequencies (ω_c): 250, 100, 50, 25 and 20 Hz (rows from 5 to 9 in Fig. 6) and applied the method to ten different trials during which the touch events occurred at different positions; proximal to distal, approximately 30-90% along the antenna. [3] found that the contact localization using the tactile sensor was accurate within the mentioned range. We found that by applying an appropriate threshold to the signal filtered with an LPF of $\omega_c = 20$ or 25 Hz, we could successfully extract the true touch events from the rest for all contact distances. All together, with a cut-off frequency of 20 Hz, the model was able to distinguish 34 true *touch* events from 35 other *disturbance*

events with an accuracy of 100%. Table I summarizes the number of false predictions in each trial for LPFs with cut-off frequencies: 100, 50, 20 and 10 Hz. In the first three cases, true touch events were correctly identified as touch events, i.e., no false negatives. However, for lower ω_c , false negatives are possible due to higher smoothing effect. For instance, at 10 Hz, all together six touch events were unidentified or missed. On the other hand, in the first three cases, some of the disturbances were misclassified as touch events and they were considered as false positive predictions. Thus, ω_c of the LPF has a crucial role for the performance. Since the decay time of the antennal oscillations is known to depend on the contact position along the antenna [3], it may be better to use an adaptive ω_c for filtering instead of a global fixed value, especially if touch events occurred along the entire length of the antenna.

The model could be further improved using a modular ESN. Since the signal to be predicted (S2) has both fast and slow dynamics (see Fig. 2), finding a single SR value that is equally well suited for different frequency components is difficult. However, this is crucial for the prediction quality. Generally, the SR should be small for the fast dynamics and large for the slow dynamics. Thus a modular ESN comprising a combination of local expert ESNs with different SR values and one special gating ESN could be beneficial [13]. The local expert ESNs then could test the set prediction, while the gating ESN could determine which local expert ESN had the best prediction accuracy for a given time step of the testing set. In [14], an ESN was used for direct control of the movements of antennae of a simulated hexapod. It will be interesting to use the same network as the forward model since it has dynamics related to movements of the antenna. Rich dynamics of the reservoir would surely increase the prediction capability of the forward model.

IV. CONCLUSION

In this paper, we present an Echo-State-Network-based forward model to predict the anticipated reading of a artificial tactile sensor mounted on to a wheeled robot. In fact, to our knowledge, this is the first use of ESNs in tactile sensing. The inputs to the ESN are efference copy of the motor command and a local proprioceptor reading. As a proof-of-concept, using real data but in offline, we show that the model can successfully be used to detect a tactile contact on the antenna when the robot is moving along a path with obstacles. In contrast to [3], where state-dependent modulation of input signals were used to suppress the detrimental effects of robot velocity on the antennal dynamics, here the predicted time series for the sensor due to the ego-motion is used for the suppression. The method described in this study provide an intelligent means of eliminating self-stimulation of sensors and such ESN based forward model could possibly be used for other sensory modalities. The next steps of the project will focus on:

- **Online Implementation** - Currently, the model is implemented in Matlab and the evaluation was done on sampled and stored data but in offline. We have not yet

validate the model in real situation. Hence we are aiming to port the application into c++ for fast computation and drive the whole system in real time with ongoing detection.

- **Spiked Wheel** - The next version of our robotic base will have spiked or toothed wheels. This can emulate touch-down of a *walking* legged robot and will produce reliable data for self stimulation of the sensors due to impacts caused by the walking.
- **3D Antennal Movements** - In the present study, for simplicity, the antenna was set to oscillate only in a vertical plane and the robot was moving in forward direction, i.e., no active turning was involved. In 3D exploration task, the two components (X, Y) of the sensors' will have individual importance and the use of combined signal (Z) may not be appropriate. Additionally, the event detection model will have to take into account the direction of the antennal contact as well.

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